

DATABASE LAB-2

Task 1

The table violates **1NF**. The seat_numbers column stores a comma-separated list (e.g. "12A, 12B"), so its values are not atomic. This makes it impossible to tell which seat belongs to which passenger, or to search or validate individual seats.

Once the table is in 1NF, it also violates:

- **2NF**: partial dependencies on part of the composite key.
passenger_full_name depends only on the passport number, and the flight data depends only on the booking.
- **3NF**: transitive dependencies such as flight_number → departure_airport_name → departure_city.

Task 2

Step 1: 1NF

Each seat goes in its own row, so one row means one passenger in one booking.

Booking_receipt_1NF(booking_id (PK), passenger_passport_number (PK), passenger full name, seat number, flight number, departure airport name, departure city, arrival airport name, arrival city, airline name, ticket price)

The primary key is the composite (booking_id, passenger_passport_number).

Step 2: 2NF

Partial dependencies (attributes that depend on only part of the key) are removed:

- Passenger(passport_number (PK), full_name)
- Booking(booking_id (PK), flight_number (FK → Flight), ticket_price)

- Booking_passenger(booking_id (PK, FK → Booking), passport_number (PK, FK → Passenger), seat_number)
- Flight(flight_number (PK), departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name)

Step 3: 3NF

Transitive dependencies (flight_number → departure_airport_name → departure_city, and the same for arrival) are removed, and airlines are moved into their own table:

- Airport(airport_id (PK), airport_name, city)
- Airline(airline_id (PK), airline_name)
- Flight(flight_number (PK), departure_airport_id (FK → Airport), arrival_airport_id (FK → Airport), airline_id (FK → Airline))
- Passenger(passport_number (PK), full_name)
- Booking(booking_id (PK), flight_number (FK → Flight), ticket_price)
- Booking_passenger(booking_id (PK, FK → Booking), passport_number (PK, FK → Passenger), seat_number)

Task 3

	<u>Functional dependency</u>	<u>Where it is used</u>
1	<u>booking_id,</u> <u>passenger passport number</u> → <u>seat number</u>	1NF/2NF: <u>key of</u> <u>Booking passenger</u>
2	<u>passenger passport number</u> → <u>passenger full name</u>	2NF: <u>extracting Passenger</u>
3	<u>booking_id</u> → <u>flight number,</u> <u>ticket price</u>	2NF: <u>extracting Booking</u>
4	<u>flight number</u> → <u>departure airport name,</u> <u>arrival airport name, airline name</u>	2NF: <u>extracting Flight</u>
5	<u>departure airport name</u> → <u>departure city</u>	3NF: <u>extracting Airport</u>
6	<u>arrival airport name</u> → <u>arrival city</u>	3NF: <u>extracting Airport</u>
7	<u>airline name</u> → <u>airline_id</u> (and vice versa)	3NF: <u>extracting Airline</u>

Task 4

Primary key: flight_id.

Functional dependencies:

flight_id $\rightarrow$ sch_departure_time, sch_arrival_time, departing_airport_id,
arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time,
act_arrival_time, created_at, updated_at

Conclusion: Flights is in 3NF (and also in BCNF).

- **1NF:** all values are atomic.
- **2NF:** the key is a single column, so partial dependencies are impossible.
- **3NF:** there are no transitive dependencies between non-key attributes.
departing_airport_id, arriving_airport_id and airline_id do not determine any
other column in Flights. The airport's name, city and country depend on
airport_id and are stored in the Airport table. The airline's name and country
depend on airline_id and are stored in the airline table. So a chain like
flight_id $\rightarrow$ departing_airport_id $\rightarrow$ city does not occur inside Flights. The

Task 4

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flight_id → sch_departure_time, sch_arrival_time, departing_airport_id,
arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time,
act_arrival_time, created_at, updated_at

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airport_id and are stored in the Airport table. The airline's name and country
depend on airline_id and are stored in the airline table. So a chain like
flight_id → departing_airport_id → city does not occur inside Flights. The
gates (departing_gate, arriving_gate) do not determine the airport either
(gate "A1" exists in many airports), so they depend only on the flight.

```
1 CREATE SCHEMA receipt_3nf;
2 SET search_path TO receipt_3nf;
3
4 CREATE TABLE Airport (
5     airport_id INT NOT NULL PRIMARY KEY,
6     airport_name VARCHAR(100) NOT NULL,
7     city VARCHAR(50) NOT NULL
8 );
9
10 CREATE TABLE Airline (
11     airline_id INT NOT NULL PRIMARY KEY,
12     airline_name VARCHAR(50) NOT NULL
13 );
14
15 CREATE TABLE Flight (
16     flight_number VARCHAR(10) NOT NULL PRIMARY KEY,
17     departure_airport_id INT NOT NULL REFERENCES Airport(airport_id),
18     arrival_airport_id INT NOT NULL REFERENCES Airport(airport_id),
19     airline_id INT NOT NULL REFERENCES Airline(airline_id)
20 );
21
22 CREATE TABLE Passenger (
23     passport_number VARCHAR(20) NOT NULL PRIMARY KEY,
24     full_name VARCHAR(100) NOT NULL
25 );
26
27 CREATE TABLE Booking (
28     booking_id INT NOT NULL PRIMARY KEY,
29     flight_number VARCHAR(10) NOT NULL REFERENCES Flight(flight_number),
30     ticket_price DECIMAL(7,2) NOT NULL
31 );
32
33 CREATE TABLE Booking_passenger (
34     booking_id INT NOT NULL REFERENCES Booking(booking_id),
35     passport_number VARCHAR(20) NOT NULL REFERENCES Passenger(passport_number),
36     seat_number VARCHAR(5) NOT NULL,
37     PRIMARY KEY (booking_id, passport_number)
38 );
```

```
37 | PRIMARY KEY (booking_id, passport_number)
38 | );
39
40
41 INSERT INTO Airport (airport_id, airport_name, city) VALUES
42 (1, 'Almaty International', 'Almaty'),
43 (2, 'Astana International', 'Astana'),
44 (3, 'Istanbul Airport', 'Istanbul'),
45 (4, 'Sheremetyevo', 'Moscow'),
46 (5, 'Dubai International', 'Dubai');
47
48 INSERT INTO Airline (airline_id, airline_name) VALUES
49 (1, 'Air Astana'),
50 (2, 'Turkish Airlines'),
51 (3, 'Aeroflot'),
52 (4, 'flydubai'),
53 (5, 'SCAT Airlines');
54
55 INSERT INTO Flight (flight_number, departure_airport_id, arrival_airport_id, airline_id) VALUES
56 ('KC101', 1, 2, 1),
57 ('KC102', 2, 1, 1),
58 ('TK351', 1, 3, 2),
59 ('SU1954', 1, 4, 3),
60 ('FZ706', 1, 5, 4),
61 ('DV761', 1, 2, 5);
62
63 INSERT INTO Passenger (passport_number, full_name) VALUES
64 ('KZ1000001', 'Aigerim Nurpeisova'),
65 ('KZ1000002', 'Dana Nurpeisova'),
66 ('KZ1000003', 'Timur Serikov'),
67 ('US2000001', 'John Smith'),
68 ('RU3000001', 'Maria Ivanova'),
69 ('RU3000002', 'Anna Petrova'),
70 ('UZ4000001', 'Ali Karimov'),
71 ('KZ1000004', 'Olga Kim');
```

```
69 ('RU3000002', 'Anna Petrova'),
70 ('UZ4000001', 'Ali Karimov'),
71 ('KZ1000004', 'Olga Kim');
72
73 INSERT INTO Booking (booking_id, flight_number, ticket_price) VALUES
74 (1, 'KC101', 85.00),
75 (2, 'KC102', 90.00),
76 (3, 'TK351', 320.00),
77 (4, 'SU1954', 280.00),
78 (5, 'KC101', 85.00),
79 (6, 'FZ706', 210.00),
80 (7, 'DV761', 60.00);
81
82 INSERT INTO Booking_passenger (booking_id, passport_number, seat_number) VALUES
83 (1, 'KZ1000001', '12A'),
84 (1, 'KZ1000002', '12B'),
85 (2, 'KZ1000003', '5C'),
86 (3, 'US2000001', '20F'),
87 (4, 'RU3000001', '8D'),
88 (5, 'RU3000002', '14A'),
89 (6, 'UZ4000001', '3A'),
90 (7, 'KZ1000004', '2B');
```



```
1  CREATE TABLE Airline_info (
2      airline_id      INT          NOT NULL PRIMARY KEY,
3      airline_code    VARCHAR(30)  NOT NULL,
4      airline_name     VARCHAR(50)  NOT NULL,
5      airline_country  VARCHAR(50)  NOT NULL,
6      created_at       TIMESTAMP    NOT NULL,
7      updated_at       TIMESTAMP    NOT NULL,
8      info             VARCHAR(50)  NOT NULL
9  );
10
11 CREATE TABLE Airport (
12     airport_id      INT          NOT NULL PRIMARY KEY,
13     airport_name    VARCHAR(50)  NOT NULL,
14     country         VARCHAR(50)  NOT NULL,
15     state           VARCHAR(50)  NOT NULL,
16     city            VARCHAR(50)  NOT NULL,
17     created_at      TIMESTAMP    NOT NULL,
18     updated_at      TIMESTAMP    NOT NULL
19 );
20
21 CREATE TABLE Passengers (
22     passenger_id     INT          NOT NULL PRIMARY KEY,
23     first_name       VARCHAR(50)  NOT NULL,
24     last_name        VARCHAR(50)  NOT NULL,
25     date_of_birth    DATE         NOT NULL,
26     gender           VARCHAR(50)  NOT NULL,
27     country_of_citizenship VARCHAR(50) NOT NULL,
28     country_of_residence VARCHAR(50) NOT NULL,
29     passport_number  VARCHAR(20)  NOT NULL,
```

```
27 country_of_citizenship VARCHAR(50) NOT NULL,
28 country_of_residence VARCHAR(50) NOT NULL,
29 passport_number VARCHAR(20) NOT NULL,
30 created_at TIMESTAMP NOT NULL,
31 updated_at TIMESTAMP NOT NULL
32 );
33
34 CREATE TABLE Flights (
35     flight_id INT NOT NULL PRIMARY KEY,
36     sch_departure_time TIMESTAMP NOT NULL,
37     sch_arrival_time TIMESTAMP NOT NULL,
38     departing_airport_id INT NOT NULL,
39     arriving_airport_id INT NOT NULL,
40     departing_gate VARCHAR(50) NOT NULL,
41     arriving_gate VARCHAR(50) NOT NULL,
42     airline_id INT NOT NULL,
43     act_departure_time TIMESTAMP NOT NULL,
44     act_arrival_time TIMESTAMP NOT NULL,
45     created_at TIMESTAMP NOT NULL,
46     updated_at TIMESTAMP NOT NULL
47 );
48
49 CREATE TABLE Booking (
50     booking_id INT NOT NULL PRIMARY KEY,
51     flight_id INT NOT NULL,
52     passenger_id INT NOT NULL,
53     booking_platform VARCHAR(50) NOT NULL,
54     created_at TIMESTAMP NOT NULL,
55     updated_at TIMESTAMP NOT NULL,
56     status VARCHAR(50) NOT NULL
```

```

53      booking_platform VARCHAR(50) NOT NULL,
54      created_at        TIMESTAMP NOT NULL,
55      updated_at        TIMESTAMP NOT NULL,
56      status            VARCHAR(50) NOT NULL,
57      price             DECIMAL(7,2) NOT NULL
58  );
59
60  CREATE TABLE Booking_flight (
61      booking_flight_id INT NOT NULL PRIMARY KEY,
62      booking_id        INT NOT NULL,
63      flight_id         INT NOT NULL,
64      created_at        TIMESTAMP NOT NULL,
65      updated_at        TIMESTAMP NOT NULL
66  );
67
68  CREATE TABLE Boarding_pass (
69      boarding_pass_id INT NOT NULL PRIMARY KEY,
70      booking_id        INT NOT NULL,
71      seat              VARCHAR(50) NOT NULL,
72      boarding_time     TIMESTAMP NOT NULL,
73      created_at        TIMESTAMP NOT NULL,
74      updated_at        TIMESTAMP NOT NULL
75  );
76
77  CREATE TABLE Baggage (
78      baggage_id INT NOT NULL PRIMARY KEY,
79      weight_in_kg DECIMAL(4,2) NOT NULL,
80      created_at  TIMESTAMP NOT NULL,
81      updated_at  TIMESTAMP NOT NULL,

```

```
80         created_at    TIMESTAMP    NOT NULL,
81         updated_at     TIMESTAMP    NOT NULL,
82         booking_id     INT          NOT NULL
83     );
84
85     CREATE TABLE Baggage_check (
86         baggage_check_id INT          NOT NULL PRIMARY KEY,
87         check_result     VARCHAR(50)  NOT NULL,
88         created_at       TIMESTAMP    NOT NULL,
89         updated_at       TIMESTAMP    NOT NULL,
90         booking_id       INT          NOT NULL,
91         passenger_id     INT          NOT NULL
92     );
93
94     CREATE TABLE Security_check (
95         security_check_id INT          NOT NULL PRIMARY KEY,
96         check_result     VARCHAR(20)  NOT NULL,
97         created_at       TIMESTAMP    NOT NULL,
98         updated_at       TIMESTAMP    NOT NULL,
99         passenger_id     INT          NOT NULL
100    );
101
102
103    -- переименовать таблицу
104    ALTER TABLE Airline_info RENAME TO airline;
105
106    -- переименовать столбец price -> ticket_price
107    ALTER TABLE Booking RENAME COLUMN price TO ticket_price;
```

```
107 ALTER TABLE Booking RENAME COLUMN price TO ticket_price;
108
109 -- изменить тип departing_gate на text
110 ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;
111
112 -- удалить столбец info
113 ALTER TABLE airline DROP COLUMN info;
114
115 -- =====
116 -- 3. СВЯЗИ (FOREIGN KEYS)
117 -- =====
118
119 -- Passengers -> Security_check, Booking, Baggage_check
120 ALTER TABLE Security_check
121     ADD CONSTRAINT fk_security_passenger
122     FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
123
124 ALTER TABLE Booking
125     ADD CONSTRAINT fk_booking_passenger
126     FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
127
128 ALTER TABLE Baggage_check
129     ADD CONSTRAINT fk_baggage_check_passenger
130     FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
131
132 -- Booking -> Baggage_check, Baggage, Boarding_pass, Booking_flight
133 ALTER TABLE Baggage_check
134     ADD CONSTRAINT fk_baggage_check_booking
135     FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
```

```
134      ADD CONSTRAINT fk_baggage_check_booking
135      FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
136
137  ALTER TABLE Baggage
138      ADD CONSTRAINT fk_baggage_booking
139      FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
140
141  ALTER TABLE Boarding_pass
142      ADD CONSTRAINT fk_boarding_pass_booking
143      FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
144
145  ALTER TABLE Booking_flight
146      ADD CONSTRAINT fk_booking_flight_booking
147      FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
148
149  -- Flights -> Booking_flight
150  ALTER TABLE Booking_flight
151      ADD CONSTRAINT fk_booking_flight_flight
152      FOREIGN KEY (flight_id) REFERENCES Flights(flight_id);
153
154  -- Airport -> Flights (вылет и прилёт)
155  ALTER TABLE Flights
156      ADD CONSTRAINT fk_flights_departing_airport
157      FOREIGN KEY (departing_airport_id) REFERENCES Airport(airport_id);
158
159  ALTER TABLE Flights
160      ADD CONSTRAINT fk_flights_arriving_airport
161      FOREIGN KEY (arriving_airport_id) REFERENCES Airport(airport_id);
162
163  -- Airport -> Flight
```

```
158
159 ALTER TABLE Flights
160     ADD CONSTRAINT fk_flights_arriving_airport
161     FOREIGN KEY (arriving_airport_id) REFERENCES Airport(airport_id);
162
163 -- Airline -> Flights
164 ALTER TABLE Flights
165     ADD CONSTRAINT fk_flights_airline
166     FOREIGN KEY (airline_id) REFERENCES airline(airline_id);
```